

Module: Ecosystem Dynamics and Energy Flow

Topic: The Carbon Cycle and Trophic Levels

1. The Carbon Cycle

The carbon cycle is the process by which carbon atoms continuously travel from the atmosphere to the Earth and then back into the atmosphere. This movement is driven by four primary biological and chemical processes:

1. **Photosynthesis:** Terrestrial and aquatic plants absorb carbon dioxide (CO_2) from the atmosphere or water and convert it into glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) using sunlight.
2. **Respiration:** Animals and plants break down glucose for energy, releasing CO_2 back into the atmosphere as a byproduct.
3. **Decomposition:** When living organisms die, decomposers (like bacteria and fungi) break down their organic matter, releasing trapped carbon into the soil or atmosphere.
4. **Combustion:** The burning of fossil fuels or organic matter (like wood) rapidly releases stored carbon back into the atmosphere as CO_2 .

2. Trophic Levels and the 10% Rule

An ecosystem's energy structure can be visualized as a pyramid divided into different feeding positions, known as **trophic levels**.

- **Producers (1st Trophic Level):** Autotrophs that make their own food through photosynthesis (e.g., plants, algae).
- **Primary Consumers (2nd Trophic Level):** Herbivores that feed directly on producers (e.g., grasshoppers, deer).
- **Secondary Consumers (3rd Trophic Level):** Carnivores or omnivores that eat primary consumers (e.g., frogs, wolves).
- **Tertiary Consumers (4th Trophic Level):** Apex predators that eat secondary consumers (e.g., hawks, sharks).

The 10% Rule of Energy Transfer: In an ecosystem, only about **10%** of the energy stored as biomass in one trophic level is passed on to the next level. The remaining 90% is lost as metabolic heat during respiration or expended during daily activities.

Numerical Example:

If producers in an ecosystem trap $10,000 \text{ kJ}$ of energy from the sun:

- Primary consumers will receive $1,000 \text{ kJ}$.
- Secondary consumers will receive 100 kJ .
- Tertiary consumers will receive only 10 kJ .

1.